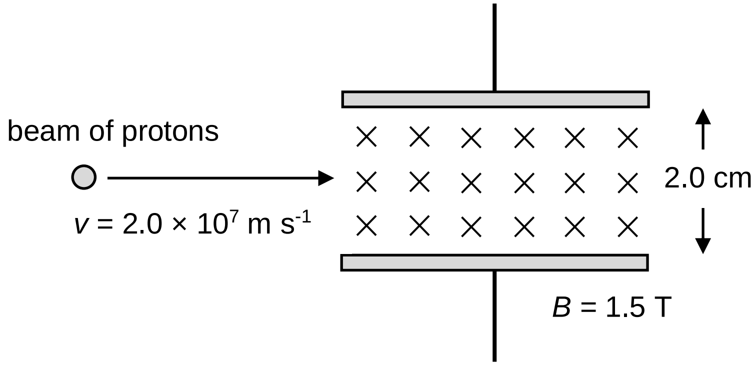
21 A beam of protons passes through a velocity selector, set up by an electric field and a magnetic field that are perpendicular to each other. The electric field is set up by a pair of charged plates with a plate separation of 2.0 cm as shown in the diagram below. The magnetic flux density, B is 1.5 T and directed into the plane of the paper.



If protons travelling at 2.0×10^{7} m s^{-1} pass through undeflected, what would be the direction and magnitude of the electric field?

	direction	magnitude
A	downwards	6.0×10^{5} N C^{-1}
B	upwards	6.0×10^{5} N C^{-1}
C	downwards	3.0×10^{7} N C^{-1}
D	upwards	3.0×10^{7} N C^{-1}

Ans: C

As proton is a positive charge, by Fleming's Left Hand rule, it will experience an upward force. To remain undeflected, the proton must also experience a downward electric force. Hence the electric field must be directed downward.

$$\text{Since } F_E = F_B$$
$$qE = Bqv$$
$$E = Bv = 1.5(2.0 \times 10^{7}) = 3.0 \times 10^{7} \text{ N C}^{-1}$$

22 A battery supplies a current of 0.025 A for 80 s. During this time it produces 18.0 J of electrical